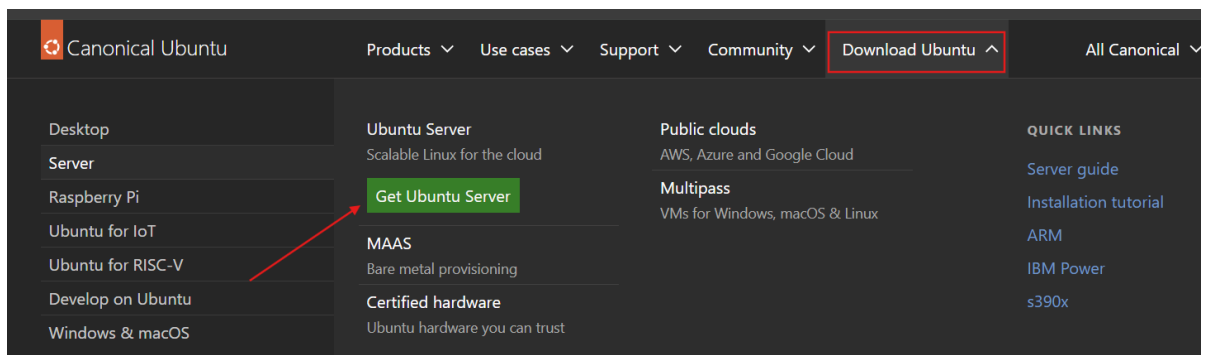
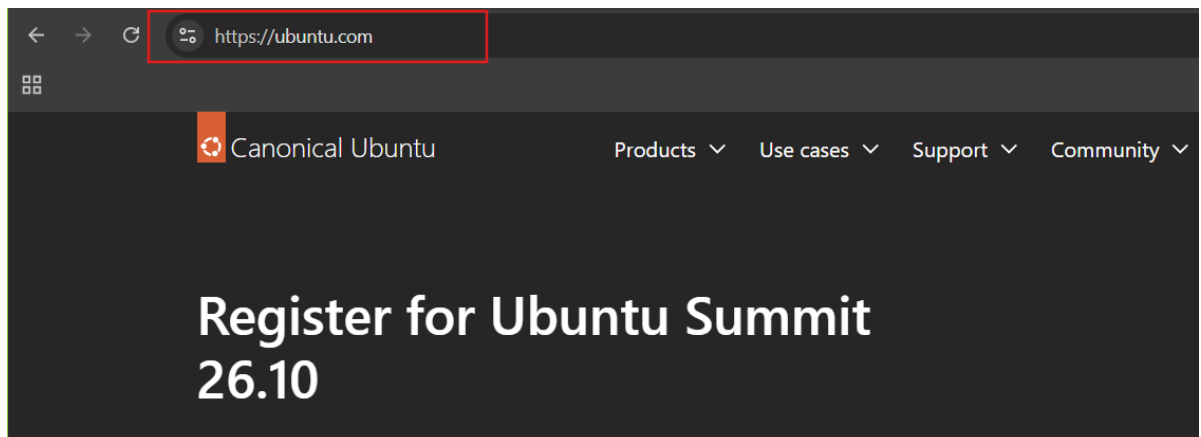


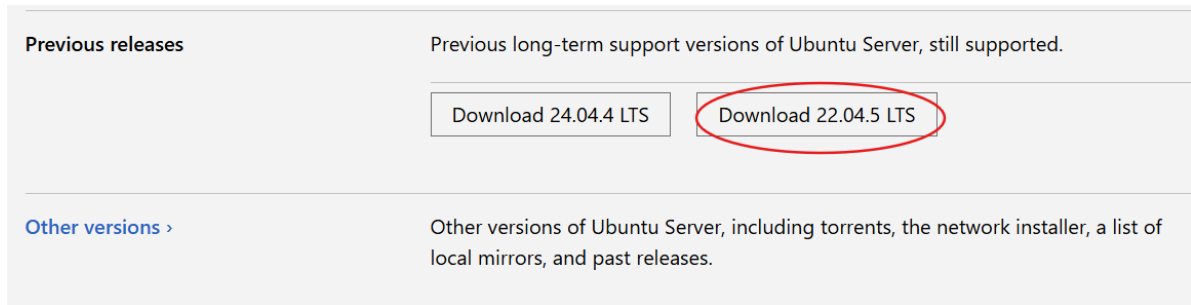
## Lab Objective

The goal of this lab is to install Ubuntu Server 22.04 on a virtual machine (VM) and enable headless SSH access from a host machine terminal.

### Step 1: Prerequisites and Downloads

- **Download VirtualBox:** Visit [virtualbox.org](https://www.virtualbox.org) and download the installer for your operating system (Windows, Mac, or Linux)
- **Download Ubuntu Server:** Visit [ubuntu.com/download](https://ubuntu.com/download) and select the Ubuntu Server option (version 22.04 is recommended) to download the ISO image





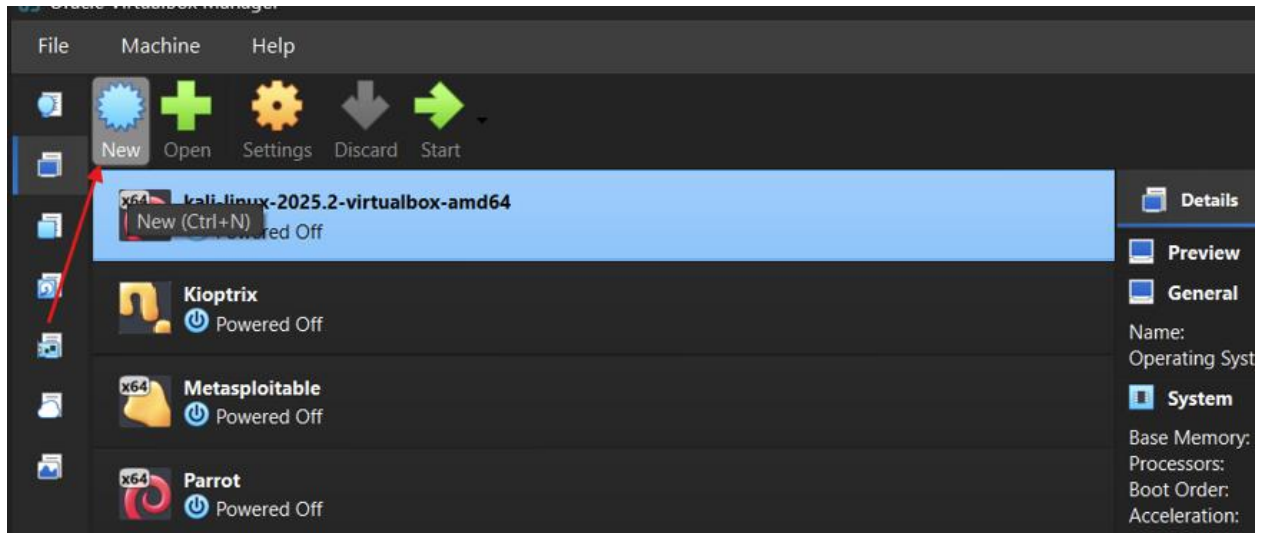
## Step 2: Install VirtualBox

Run the downloaded VirtualBox installer.

Follow the prompts to complete the installation on your host system

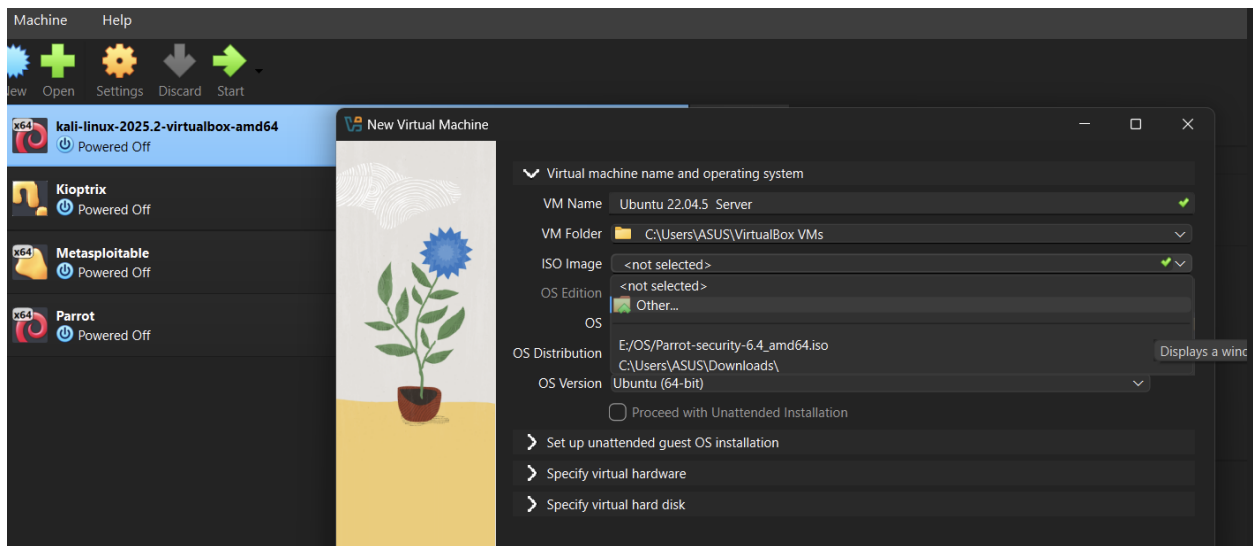
## Step 3: Create the Virtual Machine

Open VirtualBox and click the New button

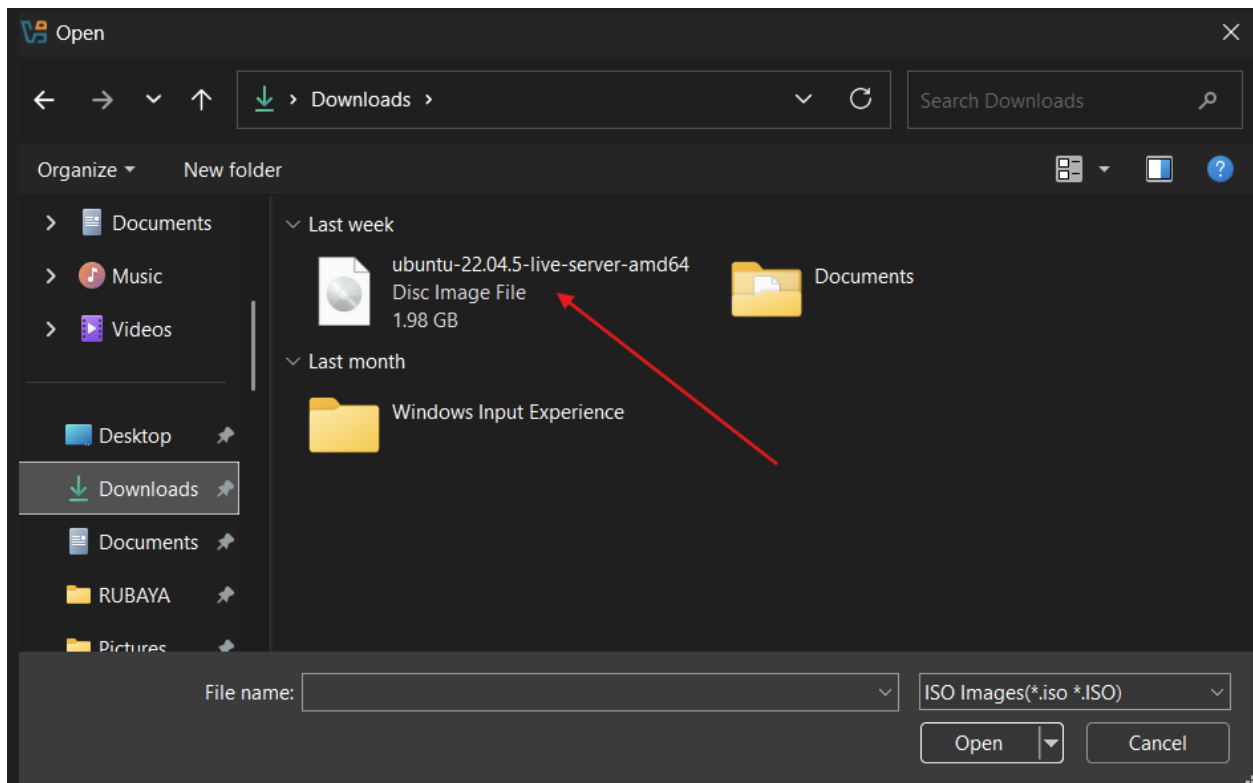


## Name and Operating System:

Name the VM (e.g., "Ubuntu 22.04 Server").



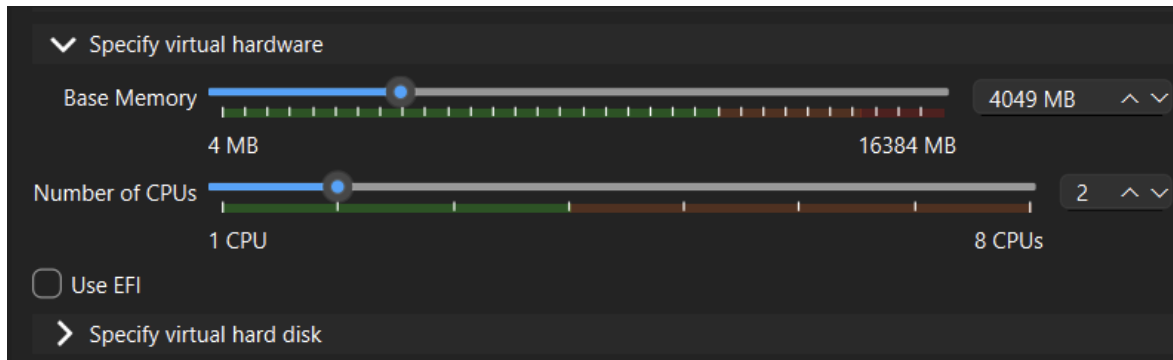
Select the downloaded Ubuntu ISO image from your computer



Check the box to Skip unattended installation to manually configure options

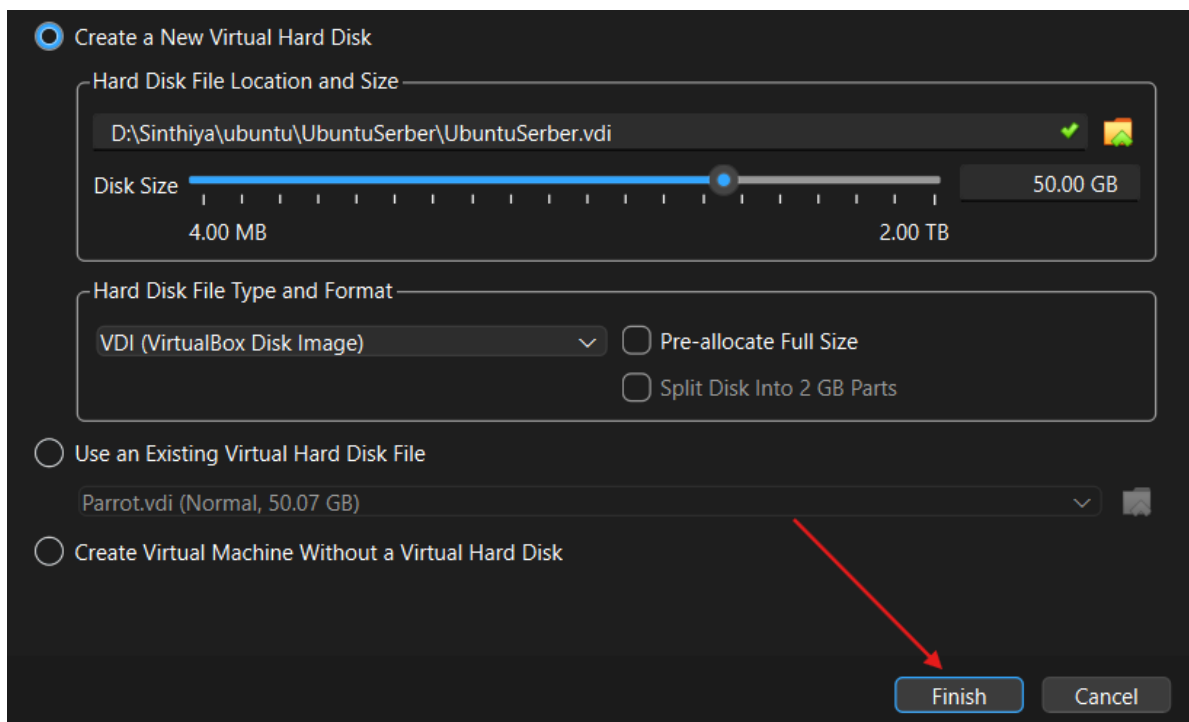
## Hardware Allocation:

- Set Base Memory to at least 4 GB (4048 MB).
- Allocate at least 2 CPU cores



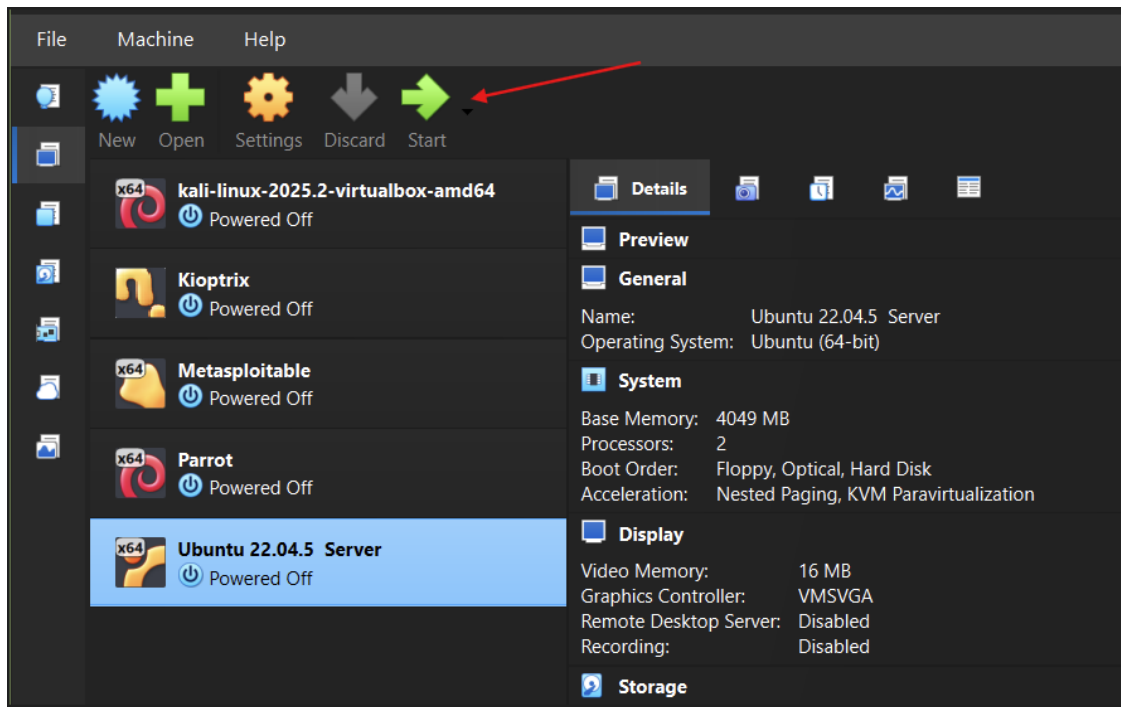
### Virtual Hard Disk:

- Create a virtual hard disk with a size of 25 GB / 30 GB / 50 GB
- Click Finish to create the VM



### Step 4: Install Ubuntu Server

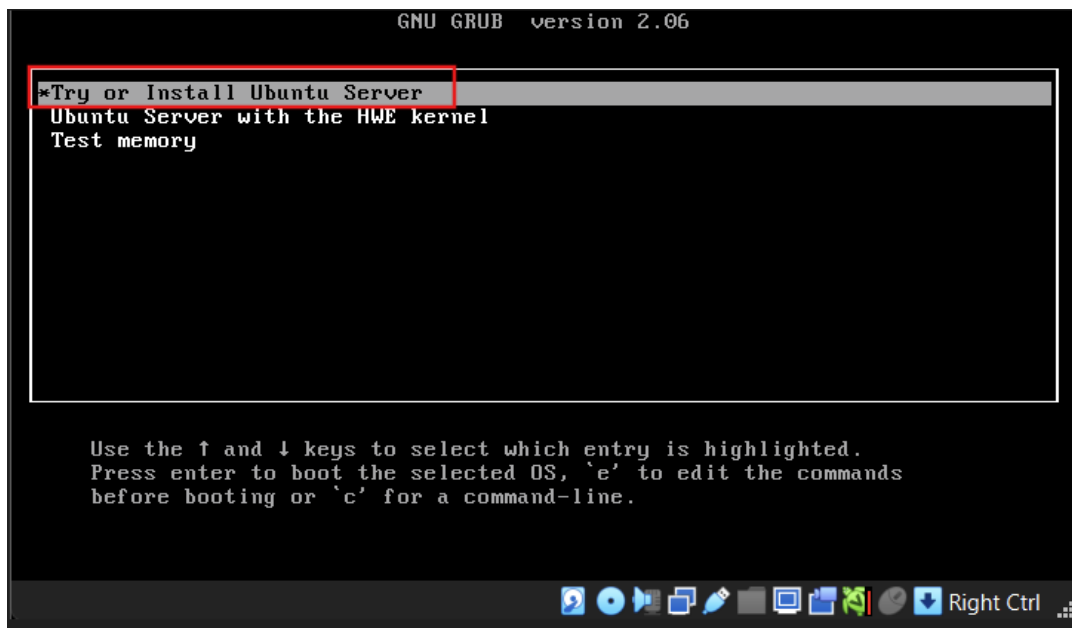
Select the VM from the list and click Start.



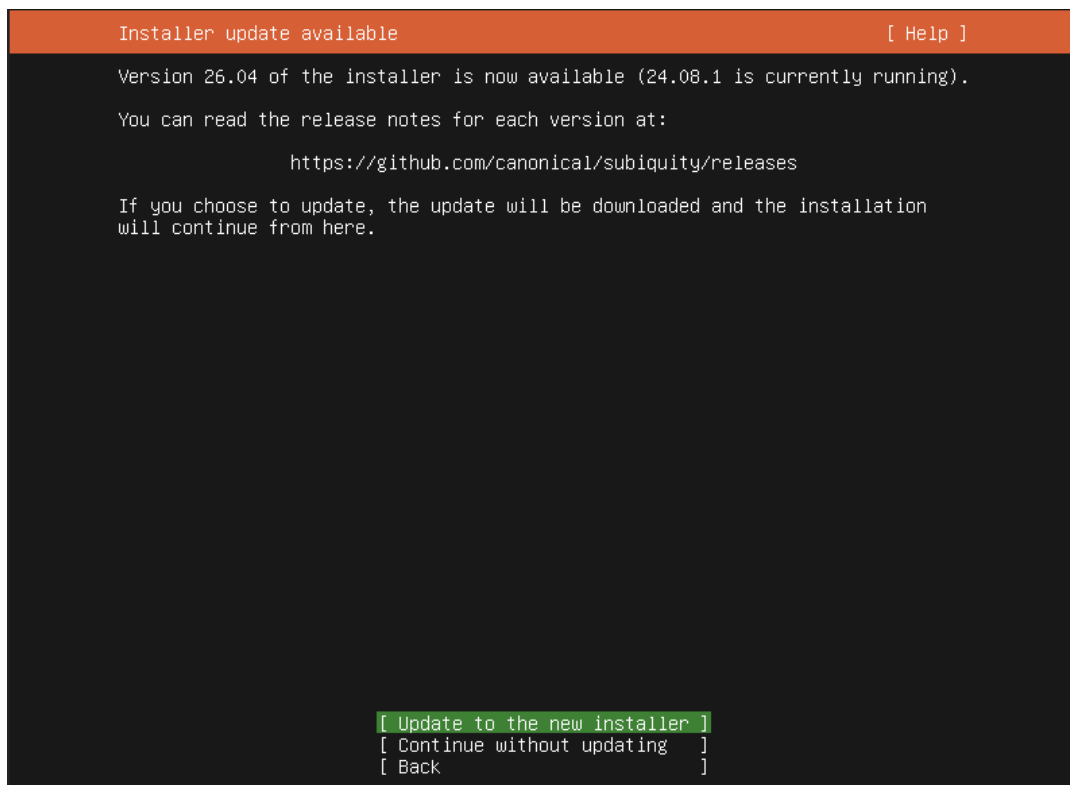
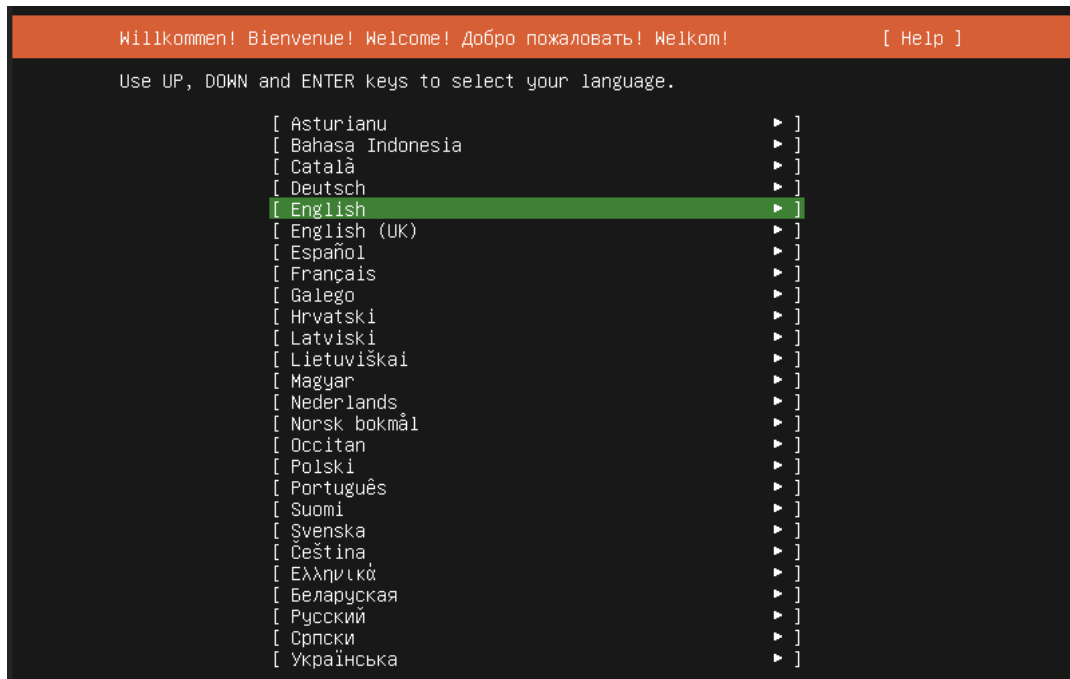
Window Management: If your mouse becomes trapped inside the VM window, press the left Command key (on Mac) or the designated host key to release it

## Installer Options:

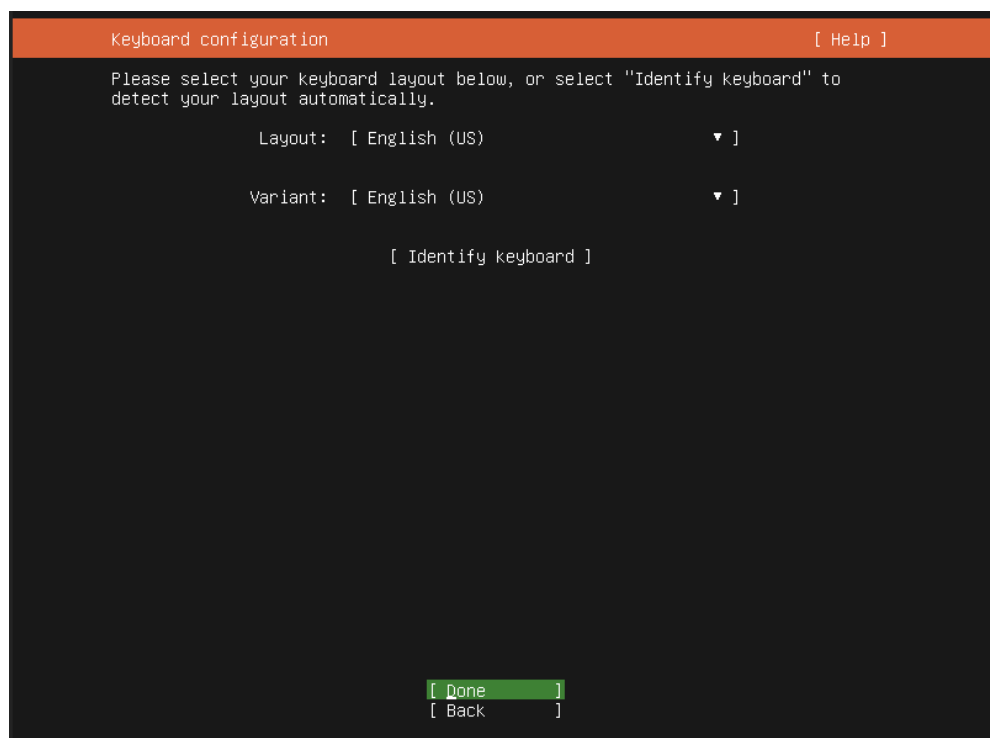
Select Try or Install Ubuntu Server



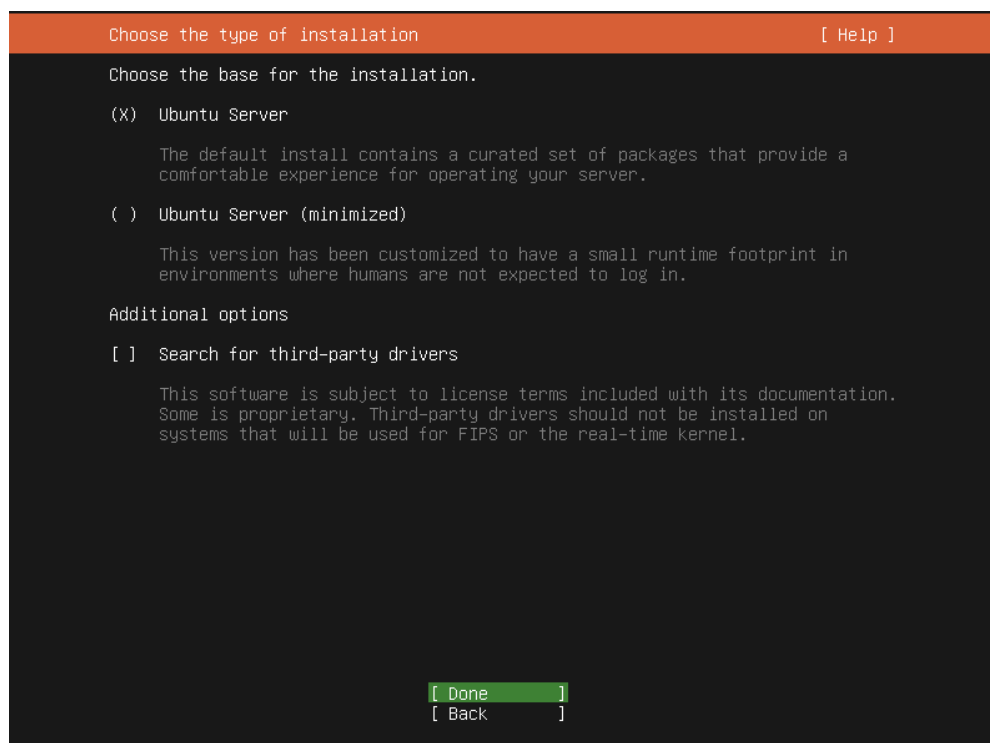
Choose your language (e.g., English) and Update to the new installer if prompted



Select your keyboard layout (e.g., English US)



## Choose the standard Ubuntu Server version



## Accept default Network and Mirror settings

Network configuration

[ Help ]

Configure at least one interface this server can use to talk to other machines, and which preferably provides sufficient access for updates.

| NAME                                                                                                      | TYPE | NOTES |
|-----------------------------------------------------------------------------------------------------------|------|-------|
| [ enp0s3                                                                                                  | eth  | - ▶ ] |
| DHCPv4 10.13.215.224/24                                                                                   |      |       |
| 08:00:27:df:43:8f / Intel Corporation / 82540EM Gigabit Ethernet Controller (PRO/1000 MT Desktop Adapter) |      |       |
| [ Create bond ▶ ]                                                                                         |      |       |

[ Done ]

[ Back ]

Proxy configuration

[ Help ]

If this system requires a proxy to connect to the internet, enter its details here.

Proxy address:

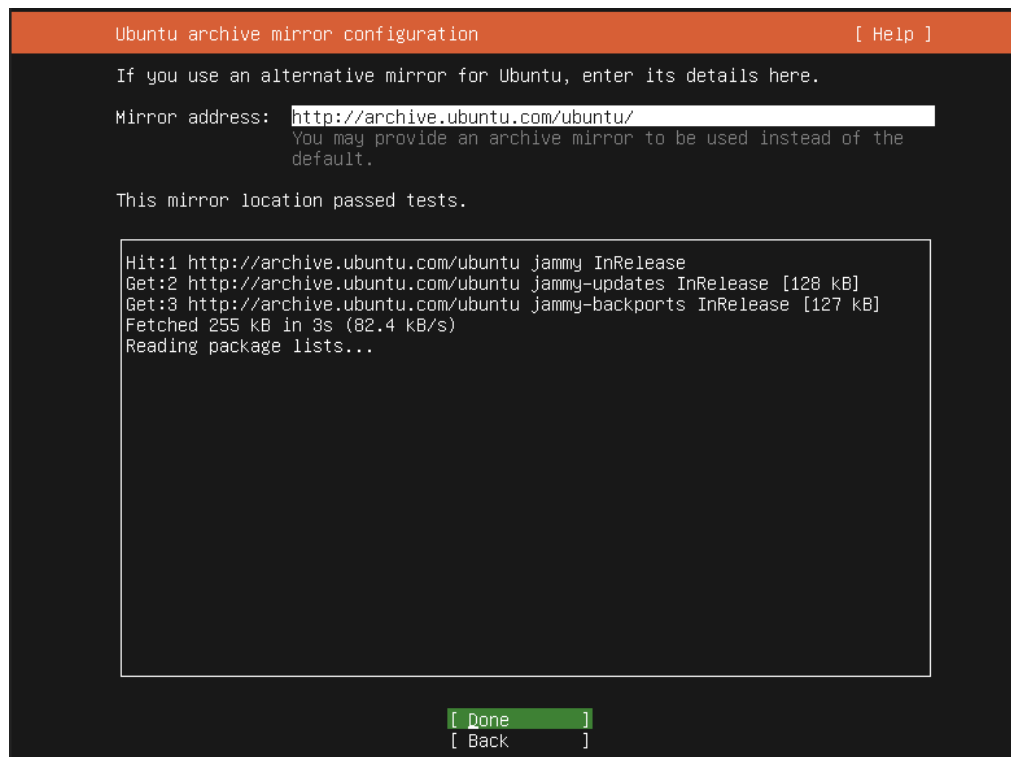
If you need to use a HTTP proxy to access the outside world, enter the proxy information here. Otherwise, leave this blank.

The proxy information should be given in the standard form of "http://[[user][:pass]@]host[:port]/".

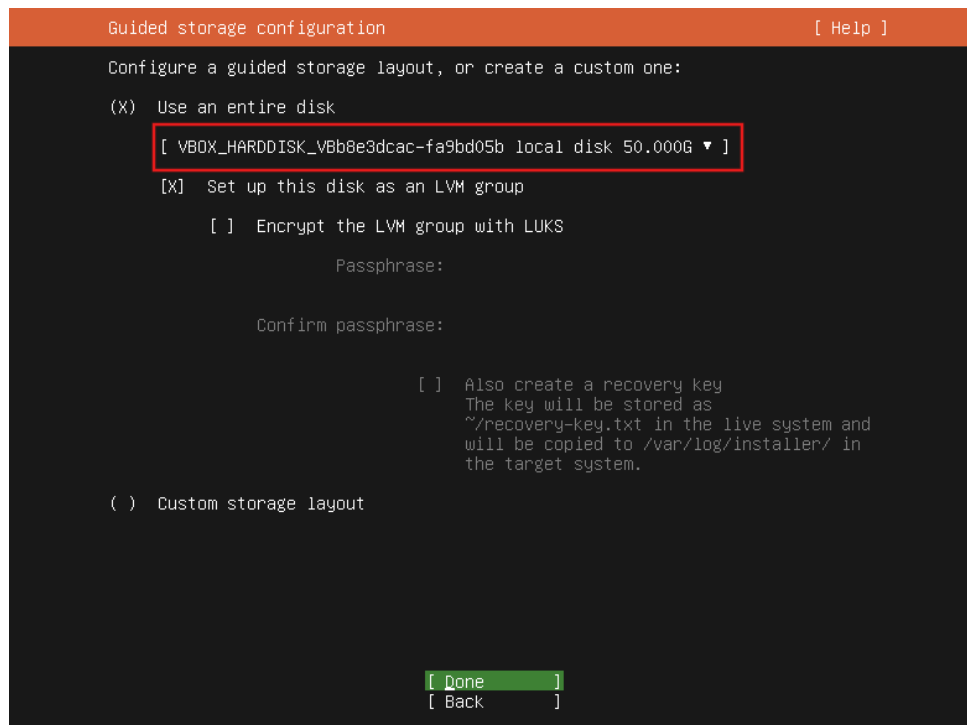
[ Done ]

[ Back ]





**Storage:** Select Use an entire disk (the 25GB /30 GB/50 GB virtual disk) and confirm the configuration



```
Storage configuration [ Help ]

FILE SYSTEM SUMMARY

MOUNT POINT  SIZE  TYPE  DEVICE TYPE
[ /           23.996G new ext4 new LVM logical volume ▶ ]
[ /boot       2.000G new ext4 new partition of local disk ▶ ]

AVAILABLE DEVICES

DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                     LVM volume group                   47.996G ▶ ]
free space                             24.000G ▶ ]

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]

USED DEVICES

DEVICE                                TYPE                                SIZE
[ ubuntu-vg (new)                     LVM volume group                   47.996G ▶ ]
ubuntu-lv    new, to be formatted as ext4, mounted at / 23.996G ▶ ]

[ VBOX_HARDDISK_VBb8e3dcac-fa9bd05b    local disk                           50.000G ▶ ]
partition 1  new, BIOS grub spacer      1.000M ▶ ]
partition 2  new, to be formatted as ext4, mounted at /boot 2.000G ▶ ]
partition 3  new, PV of LVM volume group ubuntu-vg          47.997G ▶ ]

[ Done ]
[ Reset ]
[ Back ]
```

```
Storage configuration [ Help ]

FILE SYSTEM SUMMARY

MOUNT POINT  SIZE  TYPE  DEVICE TYPE
[ /           23.996G new ext4 new LVM logical volume ▶ ]
[ /boot       2.000G new ext4 new partition of local disk ▶ ]

AVAILABLE DEVICES

Confirm destructive action

Selecting Continue below will begin the installation process and
result in the loss of data on the disks selected to be formatted.

You will not be able to return to this or a previous screen once the
installation has started.

Are you sure you want to continue?

[ No ]
[ Continue ]

partition 2  new, to be formatted as ext4, mounted at /boot 2.000G ▶
partition 3  new, PV of LVM volume group ubuntu-vg          47.997G ▶

[ Done ]
[ Reset ]
[ Back ]
```

## Profile Setup:

Enter your name, server name, username, and a secure password

Profile configuration [ Help ]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:   
The name it uses when it talks to other computers.

Pick a username:

Choose a password:

Confirm your password:

[ Done ]

## SSH Configuration:

**Crucial:** Select the option to Install OpenSSH server. This is required for remote access

Upgrade to Ubuntu Pro [ Help ]

Upgrade this machine to Ubuntu Pro for security updates on a much wider range of packages, until 2032. Assists with FedRAMP, FIPS, STIG, HIPAA and other compliance or hardening requirements.

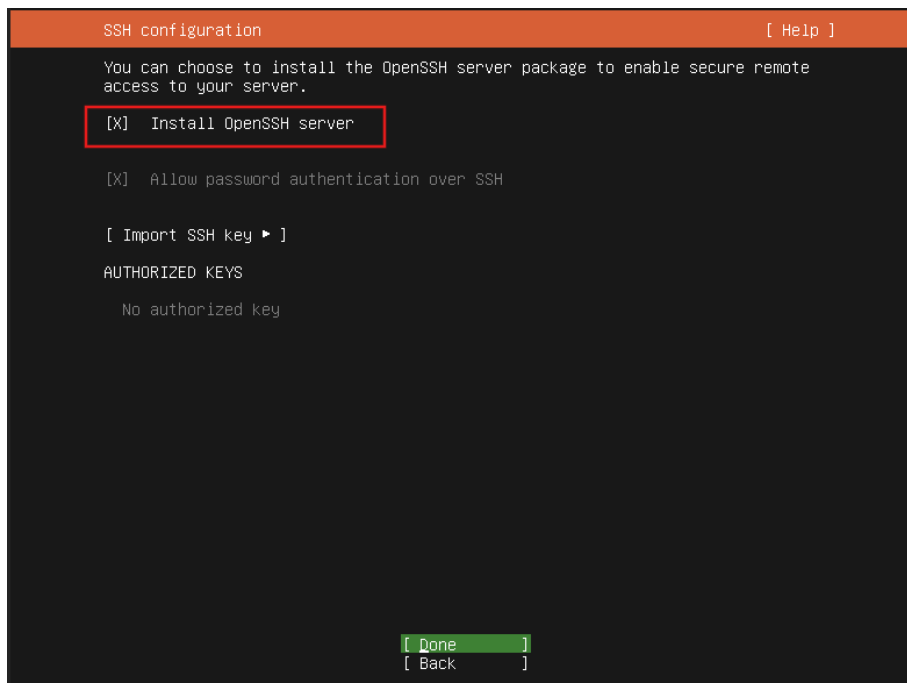
[ About Ubuntu Pro ► ]

☐ Enable Ubuntu Pro

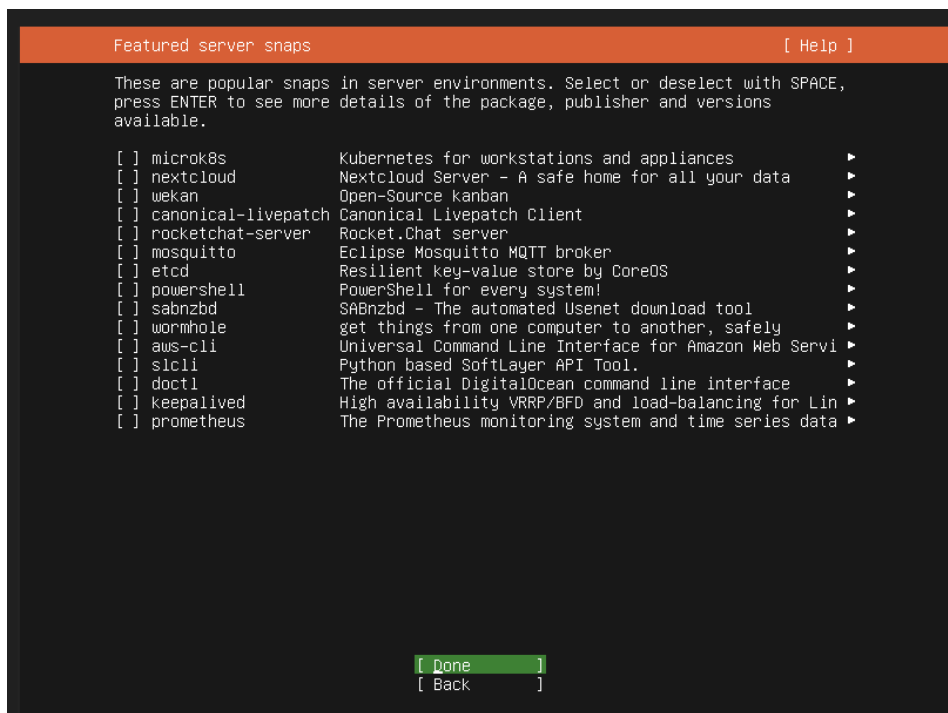
☒ Skip for now

You can always enable Ubuntu Pro later using the 'pro attach' command.

[ Continue ]  
[ Back ]



Complete Installation: Skip additional packages and wait for the installation to finish. Select Reboot Now once complete



Installing system

[ Help ]

```
running 'curtin block-meta simple'
curtin command block-meta
  removing previous storage devices
  configuring disk: disk-sda
  configuring partition: partition-0
  configuring partition: partition-1
  configuring format: format-0
  configuring partition: partition-2
  configuring lvm_volgroup: lvm_volgroup-0
  configuring lvm_partition: lvm_partition-0
  configuring format: format-1
  configuring mount: mount-1
  configuring mount: mount-0
executing curtin install extract step
curtin command install
  writing install sources to disk
  running 'curtin extract'
  curtin command extract
    acquiring and extracting image from cp:///tmp/tmpy3eer3e9/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt configuring apt
    installing missing packages \
    Installing packages on target system: ['grub-pc'] \
```

[ View full log ]

Installation complete!

[ Help ]

```
Installing packages on target system: ['grub-pc']
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
updating initramfs configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
installing openssh-server
  retrieving openssh-server
  curtin command system-install
  unpacking openssh-server
  curtin command system-install
configuring cloud-init
downloading and installing security updates
  curtin command in-target
restoring apt configuration
  curtin command in-target
subiquity/Late/run:
```

[ View full log ]

[ Reboot Now ]

```

-----BEGIN SSH HOST KEY KEYS-----
ecdsa-sha2-nistp256 AAAAE2VjZHNhLXNoYTItbmlzdHAyNTYAAAAIbmlzdHAyNTYAAABBBBCFcTNYTEfpwBV2l+zidxeNr4qY
N2N13c44W1LbWpttp2JGUksd70q11fDtlb1hpoX/W6AGgpHCxxj1rqHdLN4= root@uubuntuserver
ssh-ed25519 AAAAC3NzaC112DI1NTE5AAAAIN1iRoc0HcVVR2SwimY65VGLY1csg/Dkc0Y6o3rX9m1r root@uubuntuserver
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgQDD9ue/Qu1f50IB6B7cZ1Ji7EcZD+wPKJmHPUJCAMcYuakfQj4L+JzXfnzTPsZq
JdRI0vabiMnU0kc/0Ena59Rw7fzRppMurCbGzudP4KH7xyUKIBI0h+hshVlnwfrFkMQUR31wq/NAKXoG+D58IkNo+GRT8YNwYvNz
WI3d26ito91G1JcCu0Z0LvSotdgPVjDLLIfAmXHNGjZC5iouWi0A1xNQ+e1TtNpggQX/Giqcvg700m1Mkc3q54udvWczMnuPyZ/
qN591B4njWkme/saBNseeWfuRiTL91TXnSW3wbkpegm98E/Tx8/u55JKTPWL72Lk3FNMQ9ZVLvHkKULDAauuRfb7f1DQhX14n1NZ
kpl8jX/6dinsIHv18fgAaIqUkmjIdfkA1qmf+yUTFZFHqxyCFfPff1qKHekQM1IrM5xBG6B78WUvJLU0TaBux1HRsIgmBwbbac
LZF9QScpuc1iNKU1WxCnjJZcVGForhXsxbjMD3f8tPFxaz8BBhM= root@uubuntuserver
-----END SSH HOST KEY KEYS-----
[ 55.003725] cloud-init[1645]: Cloud-init v. 25.1.4-0ubuntu0~22.04.1 finished at Sun, 17 May 2026
10:04:38 +0000. DataSource DataSourceNone. Up 54.98 seconds
[ OK ] Finished Cloud-init: Final Stage.
[ OK ] Reached target Cloud-init target.

uubuntuserver login: rubaya
Password:

```

```

rubaya@uubuntuserver:~$
rubaya@uubuntuserver:~$
rubaya@uubuntuserver:~$
rubaya@uubuntuserver:~$
rubaya@uubuntuserver:~$
rubaya@uubuntuserver:~$
rubaya@uubuntuserver:~$
rubaya@uubuntuserver:~$

```

## Step 5: Configure SSH Port Forwarding

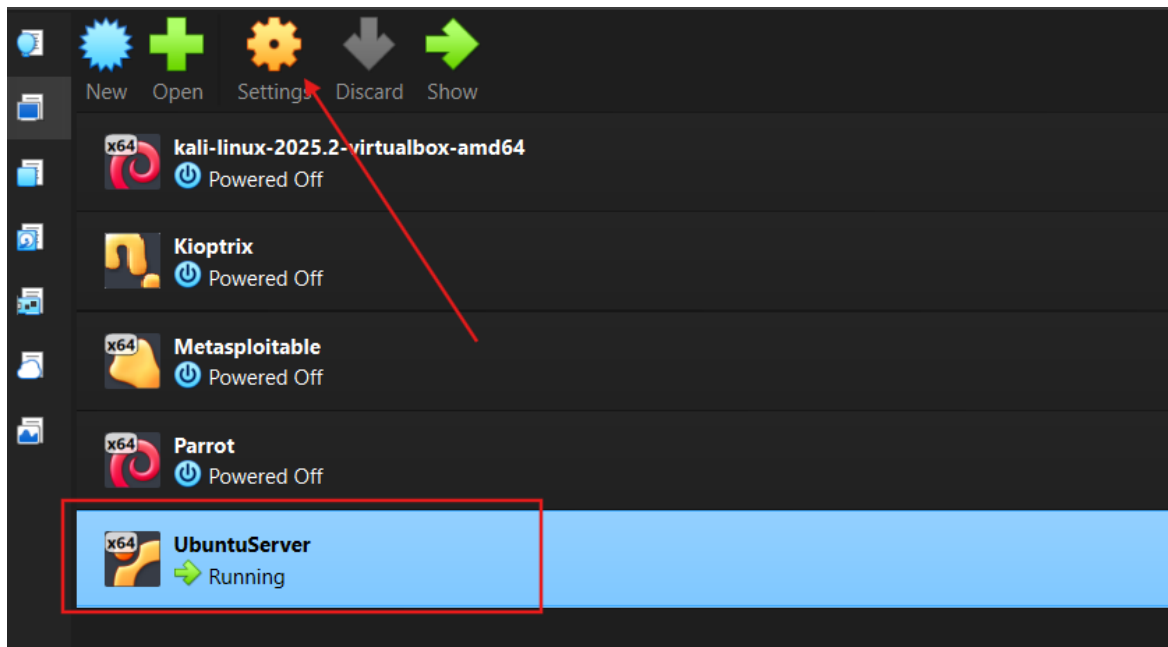
After the VM reboots and you reach the login prompt, log in once with your credentials to ensure it works

```

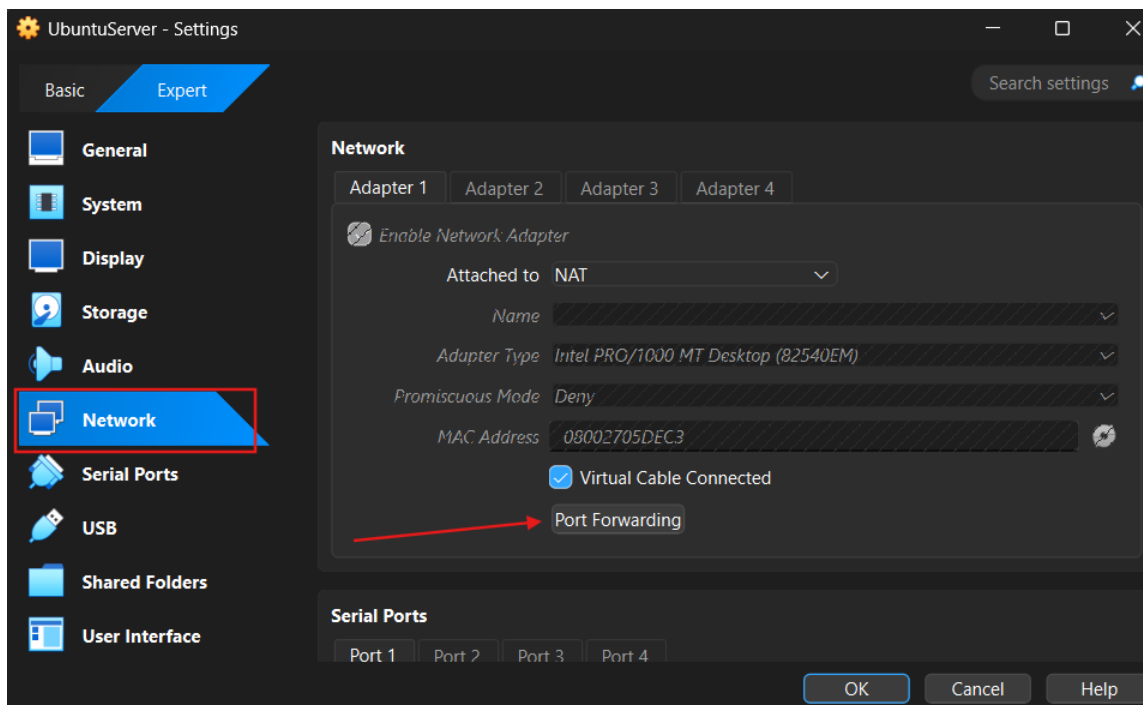
rubaya@uubuntuserver:~$ sudo apt install openssh-server -y
[sudo] password for rubaya:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssh-server is already the newest version (1:8.9p1-3ubuntu0.15).
0 upgraded, 0 newly installed, 0 to remove and 68 not upgraded.

```

Go to the VM Settings in the VirtualBox Manager



Select Network Option And go to Port Forwarding Option  
(Navigate to Network > Advanced > Port Forwarding)

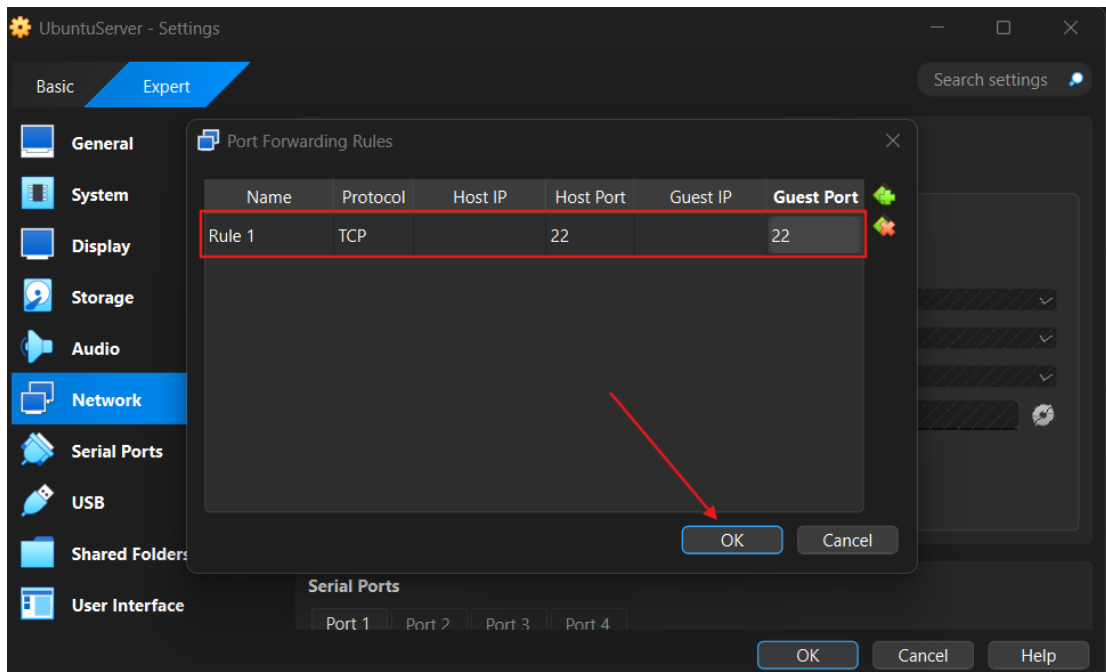
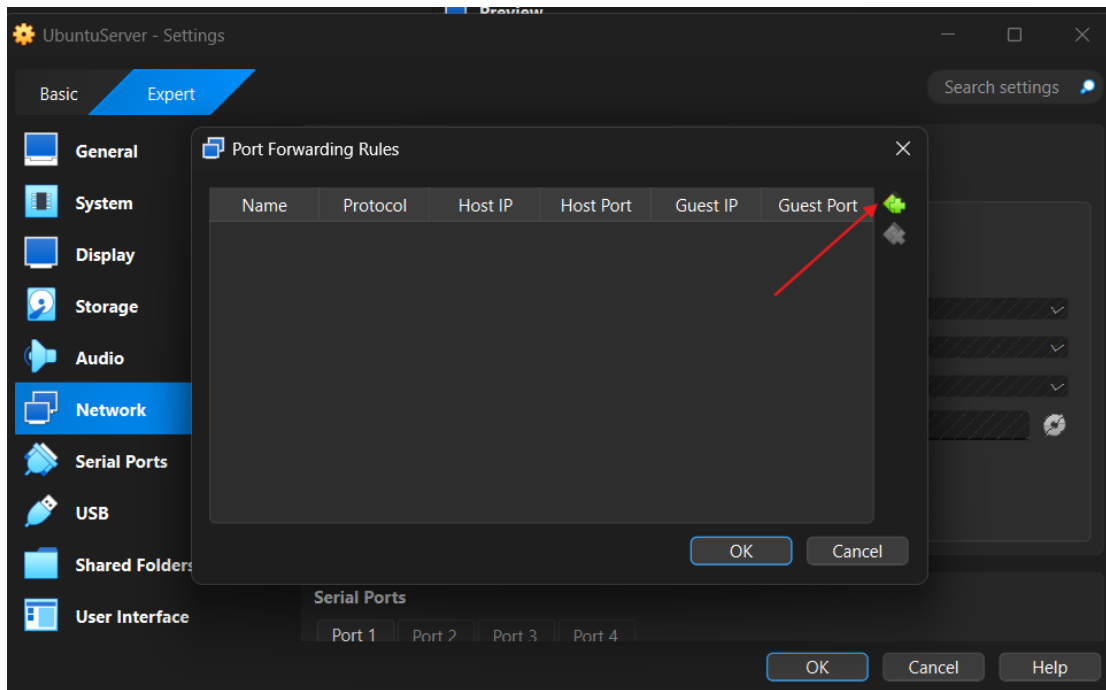


Add a new rule:

Protocol: TCP

Host Port: 22

Guest Port: 22





Click OK to save the rule.

## Step 6: Access the VM via SSH

- Open a terminal (Mac/Linux) or Command Prompt (Windows) on your host machine
- Connect to the VM using the command: **ssh username@localhost** (replace "username" with the one you created)

```
C:\Users\ASUS>ssh rubaya@localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is: SHA256:fQbHay5jlnwIsFOD8LDofGkD60whSk0AGJIQWQSnSRM
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
rubaya@localhost's password:
```

- When prompted, type yes to trust the connection and enter your VM password

```
C:\Users\ASUS>ssh rubaya@localhost
rubaya@localhost's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-179-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun May 17 11:08:56 AM UTC 2026

System load:          0.08
Usage of /:            30.7% of 22.74GB
Memory usage:         6%
Swap usage:           0%
Processes:            129
Users logged in:      1
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe05:dec3
```

```
Last login: Sun May 17 10:25:17 2026
rubaya@uubuntuserver:~$
```

Now , SSH successfully connected on Ubuntu Server

